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Effects of problem presentation durations on arithmetic strategies: a study in young and older adults

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ABSTRACT
Young and older adults were asked toaccomplish a computational estimation task (i.e. provide approximate sums to two-digit addition problems like 38 + 74) under a short versusa longer presentation duration condition. In a choice condition, participants selected the better strategy more often under a long than undera short presentation duration and after an easier than after a harder problem. In the no-choice condition, influence of presentation durationon strategy execution depended on the preceding problems, as participants showed sequence poorer strategy effects. That is, participantswere slower while executing the poorer than while executing the better strategy on current problems following better strategy problems (butequally fast after poorer strategy problems) only under short presentation duration condition. These findings have important theoreticalimplications, as current assumptions of computational models of strategies cannot explain how problem presentation durations influencestrategy selection and strategy execution.

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Strategy variability is among the most crucial features of human cognition. In a wide variety of cognitive domains, both young and older adults use several strategies to accomplish cognitive tasks, and performance depends on the strategies participants use (e.g. Siegler, 2007). A strategy is defined as "a procedure or a set of procedures to achieve a higher level goal" (Lemaire & Reder, 1999, p. 365). Performance and age-related differences in cognitive performance depend on strategies. Of great importance is how participants choose among available strategies on a given item and how they execute selected strategies. The present study contributes to this issue by investigating how presentation durations of arithmetic problems influence strategy selection and strategy execution in young and older adults.

In arithmetic, the domain in which the present study was conducted, use of multiple strategies by both young and older adults has been found when participants solve different types of problems (i.e. simple as well as complex problems) and accomplish different arithmetic tasks (e.g. production or verification tasks, exact calculation or estimation tasks; e.g. Gandini, Lemaire, & Dufau, 2008; LeFevre, Greenham, & Waheed, 1993; Lemaire & Reder, 1999). Previous research on arithmetic strategies has determined that which strategy is selected on a problem and how the selected strategy is executed are influenced by a number of factors, like problem, situation, task, and participants' characteristics. For example, in computational estimation tasks, LeFevre et al. (1993) found that participants tend to use rounding-down (i.e. rounding both operands down to the closest decades and multiply rounded operands to obtain a product estimate) on small-unit digit problems like 41×72 (and do $40 \times 70 = 2800$) and to use rounding-up (i.e. rounding both operands up to the closest decades) on large-unit digit problems like 37×69 (and do $40 \times 70 = 2800$). Lemaire, Arnaud, and Lecacheur (2004) found that participants executed the rounding-down strategy more slowly under accuracy-pressure conditions than under no-pressure conditions, especially when they solved easy problems.

A few recent studies have shown that task parameters like strategy transitions and timing of problem display also influence which strategy

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participants select on each item and how they execute strategies. Examples of effects of strategy transitions include strategy perseveration, strategy-switch costs, strategy sequential difficulty effects, and sequence poorer strategy effects. Several studies found that participants tend to select the same strategy on two consecutive problems, even if they are instructed to select the better strategy on each problem and the better strategy on each successive problem is different (e.g. Lemaire & Leclère, 2014a, 2014b). Such strategy perseveration effects have been interpreted as resulting from lower processing demands associated with repeating the same strategy on two consecutive problems relative to using two different strategies. Also, Lemaire and Lecacheur found strategy-switch costs (i.e. slower performance when participants had to execute different strategies on two successive problems relative to when they were asked to repeat the same strategy; see also Ardiale, Hodzik, & Lemaire, 2012; Ardiale & Lemaire, 2012; Lemaire & Lecacheur, 2010; Luwel, Schillemans, Onghena, & Verschaffel, 2009; Schillemans, Luwel, Bulté, Onghena, & Verschaffel, 2009). Again, lower processing demands have been assumed to account for better strategy execution on strategy-repeat items as compared to strategy-switch items. Moreover, Uittenhove and collaborators have found that execution of a strategy on a given problem is less efficient if it follows execution of a harder (relative to an easier) strategy (Uittenhove & Lemaire, 2012, 2013a; Uittenhove, Poletti, Dufau, & Lemaire, 2013). Uittenhove et al. proposed that strategy sequential difficulty effects result from decreased working-memory resources following difficult strategy execution. Finally, Lemaire and Hinault (2014) found sequential modulations of poorer strategy effects that they called sequence poorer strategy effects. In poorer strategy effects, participants obtain poorer performance for problems on which they are asked to execute a poorer strategy than on problems they have to solve with a better strategy. Hinault and Lemaire found that this poorer–better strategy difference on current problems was smaller when the previous problems were solved with a poorer strategy than when the previous problems were solved with a better strategy (see also Hinault, Dufau, & Lemaire, 2014; Hinault, Lemaire, & Phillips, 2016; Hinault, Lemaire, & Touron, 2016). These sequential modulations of poorer strategy effects were explained as the result of participants preparing themselves to process poorer strategy problems after solving poorer strategy problems and not preparing themselves after solving better strategy problems. That is, after processing poorer strategy problems, participants expected that the next problem might also be poorer strategy problems. These expectations led them to focus their attention more on the cued, poorer strategy than on the most-readily available, better strategy. This led them to more quickly process poorer strategy problems and be less influenced by the most quickly activated strategy that was not the cued strategy. In contrast, after processing a better strategy, participants expected less or did not expect to be asked to solve a poorer strategy problem and thus did not prepare themselves to execute the poorer strategy. They thus needed more time to inhibit the better strategy and executed the cued, poorer strategy.

Strategy selection and strategy execution are also influenced by timing of problem display. For example, Uittenhove and Lemaire (2013b) found that participants tended to execute strategies more efficiently when the duration between participants' response and next problem display was 600 ms relative to a shorter, 300-ms duration (see Lemaire & Brun, 2014, for similar results in children). Better execution of strategies under long duration condition was interpreted as resulting from participants' processing resources having enough time to recover from the just-executed strategy. Indeed, after executing a strategy on a given problem, participants need time to recharge the cognitive system with resources necessary to execute a strategy on the next problem. With more time between participants' response and next problem presentation, participants in Lemaire and Brun's studies solved each problem with more available processing resources and could execute strategies more efficiently.

The present study aimed at investigating the role of another task parameter on young and older adults' strategy selection and strategy execution, that is, problem presentation durations. More specifically, this study aimed at determining whether strategy selection and strategy execution would be influenced by how long problems are presented. The hypothesis under test was that participants would select the better strategy on each problem more often and execute strategies more efficiently if they have more (relative to less) time to encode problems. This would occur if longer presentation durations help participants to better encode problem features that are crucial for better strategy selection (e.g. size of unit problems).

Testing the role of presentation durations on strategy selection and strategy execution was interesting for theoretical reasons. Indeed, several theories of strategies have been proposed. Computational models of strategies (e.g. Lovett & Anderson's, 1996, Adaptative Control of Thought-Rational (ACT-R) model; Lovett & Schunn's, 1999, Represent-Construct-Choose-Learn (RCCL) model; Payne, Bettman, & Johnson's, 1993, adaptive decision-maker model; Rieskamp & Otto's, 2006, Strategy Selection Learning (SSL) model; or Siegler & Arraya's, 2005, Strategy Choice and Discovery Simulation (SCADS)* model) share several core assumptions regarding strategy choices and strategy execution. First, all models proposed that participants choose strategies on a trial-by-trial basis. This means that participants do not choose a strategy to solve all problems in an experiment, with different participants using different strategies. It also means that participants do not solve the first series of problems with one strategy, the next series of problems with another strategy, and the last series of problems with yet another strategy. Thus, both between- and within-participant variability in strategy use stems from participants' choosing strategy on each problem. Second, all models of strategy choices assume that participants select the better strategy on each problem as much as they can. To achieve this end, when a new problem is presented, participants activate mental representations of available strategies, calculate relative costs/benefits of each activated strategy, and consciously or unconsciously select the strategy that works best for a given problem (in terms of best performance and lower resources required for execution). Third, all models assume that strategies including fewer and/or simpler procedures (e.g. retrieving correct solution of arithmetic problems like 4×6 directly from memory) are easier to execute than strategies including more and/or more complex procedures (e.g. adding six 4 times). These assumptions proved sufficient to account for most findings on strategy choices, strategy execution, and age-related changes in strategic variations. However, none of these models assume that problem presentation maybe a crucial parameter influencing strategy selection and strategy execution.

There are reasons for hypothesising that presentation duration may be crucial during strategy selection and strategy execution. If presentation duration influences encoding processes are crucial in strategy selection and strategy execution. First, efficient encoding processes enable the cognitive system to deeply analyse problem features. For example, efficient encoding of two-digit multiplication problems in computational estimation tasks enables participants to analyse the size of unit digits, a crucial feature for selecting the better strategy. Determining, for example, that a problem like 37×29 has two large-unit digits is a key component for accurately determining that rounding both operands up to the closest decades is a better strategy than rounding both operands down. If participants do not have sufficient time, they may not run encoding processes to completion, encode only pieces of useful or non-useful information, or simply misencode the problems, thereby building an inadequate representation of problems. For example, if participants are presented a problem like 23×48 in a computational task where they can choose between only two strategies, rounding both operands down to their closest decades or rounding both operands up to their closest decades, misencoding unit digits of each operands may lead to inaccurate better strategy selection. This could happen if participants misencoded both unit digits as smaller than 5 and, as a consequence, select the rounding-down strategy. Moreover, efficient encoding of problems may be crucial for strategy execution. If participants do not encode problem features with sufficient precision, they may have to reencode problems after selecting a strategy and during strategy execution. Such re-encoding may interfere with efficient execution of strategies. For example, if participants encoded a problem like 23×41 too superficially and selected (appropriately) the rounding-down strategy as the better strategy, during the execution of rounding-down, this participants may need to re-encode the problem to make sure that rounding-down is the better strategy. Thus, one of the main goals of manipulating presentation duration in the present study was to test the role of encoding processes on strategy selection and strategy execution.

The second goal of the present project was to determine how the influence of encoding processes on strategy selection and strategy execution changes with age during adulthood. More than two decades of research have established that young and older adults differ in strategic aspects of cognitive performance. In a wide variety of tasks and domains, young and older adults have been found to differ in strategy repertoire, strategy distribution, strategy execution, and strategy selection.

Indeed, compared to older adults, young adults use more complex strategies and a larger number of strategies, use simpler and less efficient strategies less often, execute strategies more quickly and accurately, and select the better strategy on each problem more often and systematically (see Lemaire, 2016, for an overview).

Previous aging research has also shown that strategy use and strategy execution are not influenced by the same factors or are influenced by the same factor to different extents in young and older adults. For example, Lemaire and Leclère (2014a) found that strategy repetition (i.e. tendency to select the same strategy on two consecutive trials) was larger in older than in young adults. Ardiale et al. (2012) found larger strategy-switch costs in older than in young adults (see also Lemaire & Ardiale, 2012). Note though that some factors influence strategic variations in young and older adults to the same extent. For example, Uittenhove and Lemaire (2013b) found strategy sequential difficulty effects (i.e. slower execution of strategy on current trials following a harder strategy than after an easier strategy) of comparable magnitudes in young and older adults (see also Uittenhove, Burger, Taconnat, & Lemaire, 2015). Following previous age-related differences and similarities in strategic variations and modulations of these strategic variations by different factors, we aimed at testing whether variations in problem presentation durations influence young and older adults' strategy selection and strategy execution similarly. This was expected to determine whether the role of encoding processes on strategies changes with age during adulthood.

In the present experiment, we asked young and older adults to accomplish computational estimation tasks either in a condition where problems were presented for a short duration (short presentation duration condition) or in a condition where problems were presented for a longer duration (long presentation duration condition). To assess strategy selection and strategy execution independently, we used the choice/no-choice method (Siegler & Lemaire, 1997). That is, in each presentation duration condition, participants were given a set of two-digit addition problems, the sum of which could be estimated either by the rounding-down or the rounding-up strategies. Then, they were given two other sets of (matched) problems, each of which had to be solved by one of these two rounding strategies. The general hypothesis under test was that longer duration would enable participants to more deeply analyse problem features (e.g. size of unit and decade digits) that are crucial for better strategy selection and strategy execution. This hypothesis makes several predictions that we tested here. First, participants were expected to select the better strategy more often on each problem (in the choice condition) and to execute the required strategy more efficiently (in the no-choice conditions) under the long presentation condition than under the short presentation duration condition. Second, strategy transition effects, previously documented in the literature, were expected to be modulated by presentation durations. More specifically, we expected decreased strategy sequential difficulty effects on strategy selection (i.e. differences in mean percentages of better strategy selection on problems following harder problems and after easier problems) and decreased sequence poorer strategy effects (i.e. differences in strategy performance between poorer strategy problems and better strategy problems following poorer strategy problems vs. better strategy problems). These were expected because longer problem presentation durations would enable participants' strategy selection and strategy execution on current problems to be less contaminated by interference from preceding problems.

Given well-known cognitive slowing with age (Salthouse, 1996), we expected older adults' strategy selection and strategy execution to be more influenced by problem presentation durations. Indeed, displaying problems for a longer duration was expected to help older adults, who need more time to accomplish the same task accurately, more than young adults. This hypothesis makes the following main predictions. First, increase in mean percentages of better strategy selection under long presentation duration relative to under short presentation duration was expected to be larger in older than in young adults. Second, larger decreased magnitudes both in strategy sequential difficulty effects and in sequence poorer strategy effects in longer presentation duration relative to short presentation duration were expected to be larger in older than in young adults.

Method

Participants

Ninety-two participants were tested; 46 young adults and 46 older adults voluntarily participated in this experiment. Older adults completed the Mini-

Mental State Examination (MMSE; Folstein, Folstein, & McHugh, 1975) for potential dementia screening. All individuals had scores higher than 27 (mean: 29.2). Therefore, none were excluded.

Also, participants completed a French version of the Mill-Hill Vocabulary Scale (MHVS; Deltour, 1993; Raven, 1965) to assess their verbal fluency. The MHVS consists of 33 items distributed across 3 pages. Each item was a target word followed by six proposed words, and the task consisted in identifying which of the proposed words had the same meaning as the target word. The number of correct items assessed the level of verbal fluency.

Finally, participants' arithmetic fluency was assessed with the Tempo Test Rekenen (TTR; de Vos, 1992). The TTR is a standardised paper-and-pencil test that takes five minutes. It consists of 200 arithmetic number fact problems presented in 5 rows, 4 of each operation type and 1 with mixed addition, subtraction, multiplication, and division problems. Within each row, the problems increase in difficulty. All participants were given one minute per row and were instructed to solve the problems as fast and accurately as possible. The number of correct answers on each of the rows was summed to yield a total arithmetic score. Participants' characteristics are presented in Table 1.

Stimuli

The experiment included three sets of 12 two-digit addition problems and three sets of 32 two-digit addition problems, for a total of 132 individual problems. Half the problems in each set were so-called heterogeneous problems and the other problems homogeneous problems. Heterogeneous problems included one operand with its unit digit smaller than 5 and the other with its unit digit larger than 5. Moreover, half the heterogeneous problems had the sum of their unit digits smaller than 10 (e.g. 32 + 46), so that rounding-down (e.g. doing 30 + 40 to estimate 32 + 46) was the better strategy (i.e. the strategy that yielded the closest estimates from correct sums) on these problems. The other heterogeneous problems had the sum of their unit digits larger than 10 (e.g. 32 + 49), so that rounding-up (e.g. doing 40 + 50 to estimate 32 + 49) was the better strategy on these problems. Homogenous problems were problems with the unit digit of both operands smaller than 5 (e.g. 32 + 41) or larger than 5 (e.g. 38 + 46). Half the homogeneous problems had the unit digits of both operands smaller than 5 so that participants would obtain better estimates on these problems when correctly executing the rounding-down strategy. The other homogeneous problems had the unit digits of both operands larger than 5, so that rounding-up strategy was the better strategy. Correct sums and mean per cent deviations for each type of problem and in each set were controlled so that strategy selection would not be contaminated by these factors.

Based on previous findings in arithmetic (see Cohen-Kadosh & Dowker, 2015, for an overview), we controlled the following factors: (a) no operands had a 0 unit digit (e.g. 30 + 49) or a 5 unit digit (e.g. 35 + 49), (b) no digits were repeated within operands (e.g. 33 + 49), (c) no reverse orders of operands were used (e.g. 32 + 46 and 46 + 32), (d) the first operand was larger than the second operand in half the problems (e.g. 42 + 36), and vice versa (e.g. 37 + 69), (e) no operand had its closest decade equal to 0, 10, or 100, and (f) size of exact sums and of mean per cent deviations between

Table 1. Participants' characteristics.

	Young adults	Older adults	Means	F
Long duration condition				
N (Females)	23 (18)	23 (10)	–	–
Mean age (years; months)	21;6	70;5	–	2,220.8*
Age range	18;6–29;6	61;8–77;1	–	–
Arithmetic fluency (SD)	111.6 (20.6)	123.7 (27.6)	117.7 (24.9)	2.87†
MHVS (SD)	23.2 (4.3)	25.3 (4.2)	24.2 (4.3)	2.89†
MMSE	–	29.3	–	–
Short duration condition				
N (Females)	23 (22)	23 (13)	–	–
Mean age (years; months)	20;6	70;5	–	1,622.8*
Age range	19;0–27;9	62;8–90;4	–	–
Arithmetic fluency (SD)	109.3 (20.8)	129.8 (22.1)	119.6 (23.6)	10.46*
MHVS (SD)	19.5 (4.0)	24.3 (7.0)	21.9 (6.1)	8.09*
MMSE	–	29.1	–	–

*$p < .01$.
†$p < .10$.

best estimates and correct sums did not differ significantly across conditions and blocks.

Procedure

Participants were individually tested in one session. Before taking the experiment, older adults took the MMSE. After the arithmetic problem solving task, all participants accomplished the MHVS and the TTR.

In the arithmetic problem solving task, problems were displayed horizontally in a 84-point Bold Courier New font (black colour) in the middle of a 15-inch computer screen. The software (E-Prime) controlled stimulus display and latency collection.

Participants were told that they were going to do computational estimation, explained as providing an approximate sum without calculating the exact sum. They were told to use only two rounding strategies, the rounding-down strategy (e.g. rounding both operands down to the nearest decades) or the rounding-up strategy (e.g. rounding both operands up to the nearest decades). Previous data (e.g. LeFevre et al., 1993; Lemaire et al., 2004; Uittenhove & Lemaire, 2013a) showed that these two rounding strategies were spontaneously used by both young and older adults.

After an initial practice period including 10 problems (5 with each rounding strategy), all participants had no difficulties with either rounding strategy; none of them tried to calculate the exact sum. Then, participants practiced on 8 problems in the choice condition for them to get familiarised with this condition.

All participants were first tested under a choice condition, then under a no-choice condition, and finally under another no-choice condition. They had to choose and execute the better rounding strategy (among two available strategies) on each problem of the first set of 32 problems. Then, they were tested in two no-choice conditions, the no-choice/rounding-down condition and the no-choice/rounding-up condition. They had to use the rounding-down strategy on all 32 problems in the no-choice/rounding-down condition, and only the rounding-up strategy on all problems in the no-choice/rounding-up condition. Half the participants saw the following order: choice condition, no-choice/rounding-down condition, and no-choice/rounding-up condition. The other participants saw the following order: choice, no-choice/rounding-up condition, and no-choice/rounding-down condition.

Half the participants in each age group were presented the problems with a short presentation duration and half with long presentation duration conditions. Participants were randomly assigned to the presentation duration conditions. For each of the short and long presentation duration conditions, presentation durations were individually titrated, as they were based on each individual's mean estimation times over the three blocks of 12 problems each in choice, no-choice/rounding-down, and no-choice/rounding-up conditions. Problems were presented for 50% of individual's mean estimation times in the short duration conditions and for 100% of individual's mean estimation times in the long duration conditions (see Table 2 for means, SDs, and ranges of presentation durations in each condition). Note that 50% of individual's mean estimation times was considered as short duration relative to the long duration condition (i.e. 100% of individual's mean estimation times), and was arbitrarily chosen, as other (even smaller) percentages of individual's mean estimation times may have been selected for the short presentation duration condition.

Each trial started with a 500-ms blank screen before a 400-ms ready signal displayed at the centre of the computer screen, followed by the two-digit addition problem. If the presentation duration was up before the answer was given, a blank screen was displayed. Timing of each response began when the problem appeared on the screen and ended when the experimenter pressed the space bar of the computer keyboard, the latter event occurring as soon as possible after the participants' response. Participants were asked to calculate

Table 2. Mean (and SD, range) presentation durations in each choice and no-choice conditions.

	Short presentation duration		Long presentation duration	
Conditions	Young adults	Older adults	Young adults	Older adults
Choice				
Means	2,173	2,446	4,293	4,906
SD	441	554	781	1,523
Range	1,480–3,083	1,472–3,716	3,281–6,025	3,107–9,687
No-choice rounding-down				
Means	1,387	1,590	2,666	3,320
SD	270	455	548	1,980
Range	94–2,068	919–2,809	1,897–3,890	1,683–11,538
No-choice rounding-up				
Means	1,910	1,921	3,665	3,839
SD	418	503	791	834
Range	1,260–2,598	1,325–3,352	2,584–5,595	2,678–5,740

out loud so that we could be sure of which strategy they used. On each trial, the experimenter recorded participants' response and which strategy was used.

Results

First, we examined whether young and older adults' strategy selection was influenced by the presentation duration conditions. Second, we asked whether sequential modulations of strategy selections in young and older adults' strategy selection were influenced by presentation durations. Next, we studied how strategy execution differed in the short and long presentation conditions. Finally, sequential modulations of strategy execution were analysed. In all results, unless otherwise noted, differences are significant to at least $p < .05$. All ANOVAs were re-run with verbal and arithmetic fluency scores as covariates. However, the same results were significant.

Effects of presentation durations on better strategy selection

A mixed-design ANOVA was performed on mean percentages of better strategy selection (i.e. better strategy selection was coded 1 if participants selected the better strategy on a problem and 0 otherwise), with a 2 (Age: young, older adults) × 2 (Presentation Duration: short, long) × 2 (Problem Type: homogeneous, heterogeneous problems) design, with repeated measures on the last factor (see means in Table 3).

All participants selected the best strategy more often in the long presentation duration condition than in the short presentation duration condition (85.0% vs. 80.8%; $F(1, 88) = 3.85$, MSe = 215.5, $\eta_p^2 = .04$, $p = .05$), and more often on homogeneous problems (94.4%) than on heterogeneous problems (71.4%; $F(1, 88) = 201.13$, MSe = 120.1, $\eta_p^2 = .70$). No other effects were significant (*Fs* < 2.0, *ns*).

Sequential modulations of better strategy selection

A mixed-design ANOVA was performed on mean percentages of better strategy selection with a 2 (Age: young, older adults) × 2 (Presentation duration: short, long) × 2 (Previous problem type: homogeneous, heterogeneous) × 2 (Current problem type: homogeneous, heterogeneous) design, with repeated measures on the last two factors.

All participants selected the better strategy more often in the long presentation duration condition

Table 3. Mean percentages of better strategy selections on current problems by young and older adults in the short or long presentation duration conditions while solving homogeneous and heterogeneous problems.

	Short presentation duration			Long presentation duration			Means		
Group	Homogeneous	Heterogeneous	Means	Homogeneous	Heterogeneous	Means	Homogeneous	Heterogeneous	Means
Young adults	92.8	66.4	79.6	96.2	71.3	83.7	68.9	94.5	81.7
Older adults	91.3	72.6	81.9	97.2	75.5	86.3	74.0	94.2	84.1
Means	92.0	69.5	80.8	96.7	73.4	85.0	71.4	94.4	82.9

than participants in the short duration (85.1% vs. 80.6%; $F(1, 88) = 4.11$, MSe = 461,4, $\eta_p^2 = .04$), and after homogeneous problems (85.5%) than after heterogeneous problems (80.1%; $F(1, 88) = 185.16$, MSe = 266.2, $\eta_p^2 = .07$). No other effects were significant (*Fs* < 2.0, *ns*).

Effects of presentation durations on strategy execution

All analyses of strategy execution were ran on participants' mean solution times and percentages of errors in the (rounding-up and rounding-down) no-choice conditions. Errors were coded 0 if the expected sum estimate and participants' estimate did not differ. They were coded 1 if participant's estimate differed from the expected sum, whether this error resulted from mistake in calculation (e.g. saying 70 as an estimate of 32 + 54 with the rounding-down strategy) or from using an incorrect strategy (e.g. using rounding-up on a given problem in the rounding-down, no-choice condition). Mixed-design ANOVAs were performed with the following designs: 2 (Age: young, older adults) × 2 (Presentation duration: short, long) × 2 (Strategy: rounding-down, rounding-up) × 2 (Problem type: homogeneous, heterogeneous problems), with repeated measures on the two last factors (see means in Table 4).

Participants were faster with the rounding-down strategy (2355 ms) than with the rounding-up strategy (3390 ms), $F(1, 88) = 332.51$, MSe = 296,607.8, $\eta_p^2 = .79$. Young adults made more errors (2.3%) than older adults (0.5%, $F(1, 88) = 17.66$, MSe = 15.9, $\eta_p^2 = .17$). All participants were less accurate with the rounding-up strategy than with the rounding-down strategy (2.0% vs. 0.9%; $F(1, 88) = 12.51$, MSe = 11.6, $\eta_p^2 = .12$). The marginally significant Age × Strategy interaction ($F(1, 88) = 3.25$, MSe = 11.6, $\eta_p^2 = .04$, $p = .08$) showed that this strategy difference tended to be larger in young adults (1.9%; $F(1, 44) = 8.35$, MSe = 19.8, $\eta_p^2 = .16$) than in older adults (0.6%; $F(1, 44) = 5.13$, MSe = 3.4, $\eta_p^2 = .10$). The Presentation Duration × Strategy × Problem Type was marginally significant, $F(1, 88) = 3.22$, MSe = 9.0, $\eta_p^2 = .04$, $p = .08$ (see Figure 1). Breakdown analyses in each presentation duration condition revealed that the Strategy × Problem Type interaction was significant in the short duration condition ($F(1, 44) = 4.70$, MSe = 9.0, $\eta_p^2 = .10$) but not in the long duration condition ($F < 1$, *ns*). In the long duration condition, participants made more errors while executing the rounding-up strategy than the

Table 4. Young and older adults' mean estimation times (in ms) and per cent errors on current problems in the short or long presentation duration condition while solving homogeneous versus heterogeneous problems with the rounding-down and rounding-up strategies.

Strategy	Rounding-down strategy			Rounding-up strategy			Means		
Problem type	Homogeneous problems	Heterogeneous problems	Means	Homogeneous problems	Heterogeneous problems	Means	Homogeneous problems	Heterogeneous problems	Means
Short presentation duration									
Young adults	2,349 (0.8)	2,347 (2.5)	2,348 (1.7)	3,494 (4.2)	3,503 (2.6)	3,498 (3.4)	2,921 (2.5)	2,925 (2.5)	2,923 (2.5)
Older adults	2,349 (0.0)	2,323 (0.3)	2,336 (0.1)	3,246 (0.8)	3,182 (0.5)	3,214 (0.7)	2,797 (0.4)	2,753 (0.4)	2,775 (0.4)
Means	2,349 (0.4)	2,335 (1.4)	2,342 (0.9)	3,370 (2.5)	3,343 (1.6)	3,356 (2.0)	2,859 (1.4)	2,839 (1.5)	2,849 (1.5)
Long presentation duration									
Young adults	2,244 (0.8)	2,232 (1.1)	2,238 (1.0)	3,306 (3.0)	3,315 (3.1)	3,311 (3.0)	2,775 (1.9)	2,773 (2.1)	2,774 (2.0)
Older adults	2,487 (0.5)	2,507 (0.0)	2,497 (0.3)	3,562 (0.8)	3,513 (1.1)	3,537 (1.0)	3,025 (0.7)	3,010 (0.6)	3,017 (0.6)
Means	2,366 (0.7)	2,369 (0.5)	2,367 (0.6)	3,434 (1.9)	3,414 (2.1)	3,424 (2.0)	2,900 (1.3)	2,892 (1.3)	2,896 (1.3)
Means									
Young adults	2,296 (0.8)	2,289 (1.8)	2,293 (1.3)	3,400 (3.6)	3,409 (2.8)	3,404 (3.2)	2,848 (2.2)	2,849 (2.3)	2,849 (2.3)
Older adults	2,418 (0.3)	2,415 (0.1)	2,416 (0.2)	3,404 (0.8)	3,347 (0.8)	3,376 (0.8)	2,911 (0.5)	2,881 (0.5)	2,896 (0.5)
Means	2,357 (0.5)	2,352 (1.0)	2,355 (0.8)	3,402 (2.2)	3,378 (1.8)	3,390 (2.0)	2,880 (1.4)	2,865 (1.4)	2,872 (1.4)

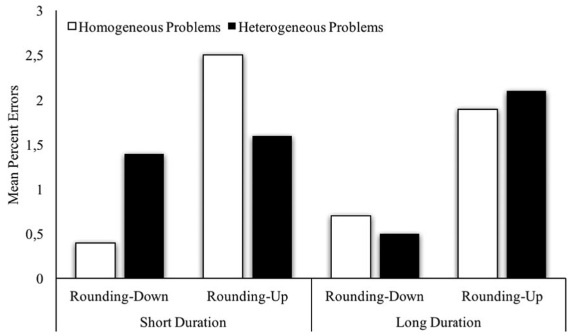

Figure 1. Mean per cent errors in each presentation duration condition for homogeneous and heterogeneous problems solved with the rounding-down and rounding-up strategies.

rounding-down strategy to solve both heterogeneous and homogeneous problems (*F*s < 5.90). In the short duration condition, participants made more errors while executing the rounding-up strategy relative to the rounding-down strategy (2.5 vs. 0.4%; $F(1, 4) = 8.26$, MSe = 12.1, $\eta_p^2 = .16$) while solving homogeneous problems but erred equally often on both strategies while solving heterogeneous problems (1.4% vs. 1.6%; $F < 1.0$). No other effects were significant either on latencies or on percentages of errors (*F*s < 3.0, *ns*).

Sequential modulations of strategy execution

Two series of ANOVAs were run. First, to analyse sequential modulations of strategy execution by the type of preceding problems, mixed-design ANOVAs were performed on participants' mean solution times and percentages of errors in the no-choice condition: 2(Age: young, older adults) × 2 (Presentation duration: short, long) × 2 (Strategy: rounding-down, rounding-up) × 2 (Previous problem type: homogeneous, heterogeneous problems), with repeated measures on the two last factors (see means in Table 5).

All participants were faster with the rounding-down strategy (2354 ms) than with the rounding-up strategy (3392 ms), $F(1, 88) = 330.59$, MSe = 299757.9, $\eta_p^2 = .79$. On percentages of errors, the main effects of age and of strategy were significant. Although participants made few errors, young adults made more errors (2.3%) than older adults (0.5%, $F(1, 88) = 17.27$, MSe = 16.3, $\eta_p^2 = .16$). All participants were less accurate with the rounding-up strategy than with the rounding-down strategy (2.0% vs. 0.8%; $F(1, 88) = 11.95$, MSe = 12.1, $\eta_p^2 = .12$).

Table 5. Young and older adults' mean estimation times (in ms) and per cent errors on current problems solved with the rounding-down and rounding-up strategies in the short or long presentation duration condition as a function of preceding problem type.

Strategy	Rounding-down			Rounding-up			Means		
Previous Problem Type	Homogeneous problems	Heterogeneous problems	Means	Homogeneous problems	Heterogeneous problems	Means	Homogeneous problems	Heterogeneous problems	Means
Short presentation duration									
Young adults	2,384 (1.7)	2,312 (1.7)	2,348 (1.7)	3,509 (2.9)	3,490 (4.0)	3,500 (3.5)	2,947 (2.3)	2,901 (2.9)	2,924 (2.6)
Older adults	2,335 (0.0)	2,333 (0.3)	2,334 (0.1)	3,249 (0.9)	3,173 (0.6)	3,211 (0.7)	2,792 (0.4)	2,753 (0.4)	2,772 (0.4)
Means	2,360 (0.9)	2,322 (1.0)	2,341 (0.9)	3,379 (1.9)	3,331 (2.3)	3,355 (2.1)	2,870 (1.4)	2,827 (1.7)	2,848 (1.5)
Long presentation duration									
Young adults	2,259 (0.3)	2,216 (1.7)	2,237 (1.0)	3,363 (2.8)	3,273 (3.0)	3,318 (2.9)	2,811 (1.6)	2,744 (2.4)	2,778 (2.0)
Older adults	2,497 (0.3)	2,497 (0.3)	2,497 (0.3)	3,558 (1.4)	3,520 (0.6)	3,539 (1.0)	3,028 (0.9)	3,009 (0.4)	3,018 (0.6)
Means	2,378 (0.3)	2,357 (1.0)	2,367 (0.6)	3,461 (2.1)	3,396 (1.8)	3,429 (2.0)	2,919 (1.2)	2,877 (1.4)	2,898 (1.3)
Means									
Young adults	2,322 (1.0)	2,264 (1.7)	2,293 (1.4)	3,436 (2.9)	3,381 (3.5)	3,409 (3.2)	2,879 (1.9)	2,823 (2.6)	2,851 (2.3)
Older adults	2,416 (0.1)	2,415 (0.3)	2,416 (0.2)	3,404 (1.2)	3,346 (0.6)	3,375 (0.9)	2,910 (0.6)	2,881 (0.4)	2,895 (0.5)
Means	2,369 (0.6)	2,339 (1.0)	2,354 (0.8)	3,420 (2.0)	3,364 (2.1)	3,392 (2.0)	2,894 (1.3)	2,852 (1.5)	2,873 (1.4)

Finally, the marginally significant Age × Strategy interaction (F(1, 88) = 2.79, MSe = 12.1, $\eta_p^2 = .03$, $p = .10$) showed that this strategy difference tended to be larger in young adults (1.9%; F(1, 44) = 7.78, MSe = 20.4, $\eta_p^2 = .15$) than in older adults (0.6%; F(1, 44) = 5.14, MSe = 3.8, $\eta_p^2 = .10$).

To determine if sequence poorer strategy effects differed in short and long presentation duration conditions, ANOVAs were performed on mean solution times and percentages of errors in the no-choice condition, with 2 (Age: young, older adults) × 2 (Presentation duration: short, long) × 2 (Previous strategy: better, poorer) × 2 (Current strategy: better, poorer) designs, with repeated measures on the two last factors (see means in Table 6).

All participants were faster when the current strategy was the better strategy than when it was the poorer strategy (2815 ms vs. 2920 ms; F(1, 88) = 33.18, MSe = 30421.3, $\eta_p^2 = .27$). The Presentation Duration × Previous Strategy × Current Strategy (F(1, 88) = 6.03, MSe = 126841.1, $\eta_p^2 = .06$) interaction was significant. This interaction showed that the Previous strategy × Current strategy interaction was significant in the short presentation duration condition (F(1, 44) = 6.37, MSe = 26029.5, $\eta_p^2 = .13$) but not in the long duration condition ($F < 1$, *ns*). In the short presentation duration condition, participants were faster on current problems estimated with the better strategy (2760 ms) than with the poorer strategy (2912 ms) after executing the better strategy on immediately preceding problems (F(1, 44) = 23.62, MSe = 22323.8, $\eta_p^2 = .35$) but were equally fast on current better and poorer strategy problems (2833 ms vs. 2865 ms; $F < 1$, *ns*) after executing the poorer strategy on the preceding problems. In the long presentation duration condition, participants were faster on current problems estimated with the better strategy than with the poorer strategy both after executing the better strategy on immediately preceding problems (2836 ms vs. 2940 ms; F(1, 44) = 10.50, MSe = 236.56, $\eta_p^2 = .19$) and after executing the poorer strategy (2831 ms vs. 2963 ms; F(1, 44) = 13.91, MSe = 28960.8, $\eta_p^2 = .24$). Analyses of mean percentages of errors revealed a main effect of age (F(1, 88) = 17.70, MSe = 15.9, $\eta_p^2 = .17$), as young adults made more errors (2.3%) than older adults (0.5%). No other effects were significant on latencies or on percentages of errors ($Fs < 2.0$, *ns*).

General discussion

Above and beyond replicating some previously reported findings (e.g. effects of strategy and problem types on participants' performance and strategy choices), this study revealed effects of problem presentation duration on participants' strategy selection and strategy execution. The present findings also showed that these effects were comparable in young and older adults. We found that both young and older adults tended to select the better strategy on each problem more often when these problems were presented for a longer (relative to a short) duration and when current problems followed homogeneous problems (for which the better strategy is easier to determine) than after heterogeneous problems (for which it is harder to determine the better strategy). Sequential modulations of better strategy selection were not influenced by presentation duration and were of comparable magnitudes in young and older adults. Moreover, under both duration presentation conditions, young and older

Table 6. Young and older adults' mean estimation times (in ms) and per cent errors on current problems solved with the better versus poorer strategy in the short or long presentation duration conditions as a function of which strategy was executed on the preceding problems.

	Previous better strategy				Previous poorer strategy			
	Current better strategy	Current poorer strategy	Differences	Means	Current better strategy	Current poorer strategy	Differences	Means
Short presentation duration								
Young adults	2,779 (2.2)	2,978 (2.9)	199 (0.6)	2,878 (2.6)	2,931 (3.0)	2,968 (2.3)	37 (−0.7)	2,950 (2.6)
Older adults	2,742 (0.3)	2,845 (0.3)	103 (0.0)	2,794 (0.3)	2,735 (1.1)	2,761 (0.0)	26 (−1.1)	2,748 (0.6)
Means	2,760 (1.3)	2,912 (1.6)	151* (0.3)	2,836 (1.4)	2,833 (2.1)	2,865 (1.1)	31 (−0.9)	2,849 (1.6)
Long presentation duration								
Young adults	2,733 (2.7)	2,843 (1.2)	110 (−1.6)	2,788 (2.0)	2,701 (1.5)	2,805 (2.5)	104 (1.0)	2,753 (2.0)
Older adults	2,939 (0.5)	3,036 (0.8)	97 (0.3)	2,988 (0.7)	2,961 (0.3)	3,122 (0.8)	160 (0.5)	3,041 (0.6)
Means	2,836 (1.6)	2,940 (1.0)	104* (−0.6)	2,888 (1.3)	2,831 (0.9)	2,963 (1.6)	132* (0.7)	2,897 (1.3)
Means								
Young adults	2,756 (2.5)	2,911 (2.0)	155 (−0.5)	2,833 (2.3)	2,816 (2.2)	2,887 (2.4)	71 (0.1)	2,851 (2.3)
Older adults	2,840 (0.4)	2,941 (0.6)	100 (0.2)	2,891 (0.5)	2,848 (0.7)	2,941 (0.4)	93 (−0.3)	2,895 (0.6)
Means	2,798 (1.4)	2,926 (1.3)	128 (−0.1)	2,862 (1.4)	2,832 (1.5)	2,914 (1.4)	82 (−0.1)	2,873 (1.4)

Note: $*p < .01$.

participants were equally fast and accurate on current problems following easier, homogeneous problems than after harder, heterogeneous problems. Finally, we found sequence poorer strategy effects on strategy execution only under short presentation duration condition in young and older adults. These findings have important implications to further our understanding of age-related differences and similarities in strategic variations underlying cognitive performance.

As predicted, presenting problems for a longer duration enabled participants to select the better strategy more often. Longer problem presentation likely enabled participants to more efficiently encode crucial problem features (e.g. size of unit digits in both operands) for constructing a clearer, more precise, and more accurate representation of problems in working memory, on the basis of which participants select the better strategy. Note that this increased better strategy selection under longer presentation duration was found for both homogeneous and heterogeneous problems. This suggests that efficient encoding processes are crucial for better strategy selection, however hard it is to select the better strategy. It was also interesting to find here that, in contrast to our prediction, longer presentation duration did not facilitate older adults' better strategy selection more than young adults'. We expected that longer presentation durations would help older adults more than young adults, especially on harder, heterogeneous problems, because older adults usually need more time to accomplish cognitive tasks, given well-known age-related decrease in processing speed (e.g. Salthouse, 1996). Note that although like in previous studies (e.g. Lemaire & Leclère, 2014a, 2014b; Lemaire et al., 2004) participants selected the better strategy more often on homogeneous than on heterogeneous problems, both age groups were very good at selecting the better strategy, even on the harder heterogeneous problems (young and older adults selected the better strategy on 70% and 73%, respectively). These percentages are higher than in some of our previous studies (e.g. young adults selected the better strategy on 61% of heterogeneous problems in Taillan, Dufau, & Lemaire, 2015; young and older adults selected the better strategy on 65% and 57%, respectively, on heterogeneous problems in Lemaire et al., 2004). Note that problem set in the present experiment included two-digit addition problems for which the better strategy was easier to determine than in our previous studies testing two-digit multiplication problems. This may have led our older adults to be less influenced by longer presentation duration than expected. This could be easily tested by selecting a problem set with problems for which the better strategy is harder to select. Also, in contrast to some previous studies of ours, our present older adults were as good as young adults at selecting the better strategy. This may be the result of (a) our present problem set including problems of insufficient difficulty, and/or (b) our sample of older adults including extremely select participants. Our older adults had very high arithmetic skills (their arithmetic fluency scores at our independent paper-and-pencil TTR test indicated higher arithmetic fluency than young adults). It is possible that older adults with lower arithmetic fluency might benefit more from longer presentation duration while trying to select the better strategy on each problem, a possibility that future studies may test.

One additional surprising result was that strategy sequential difficulty effects on better strategy selection was not modulated by presentation durations. That is, participants selected the better strategy on each problem more often after homogeneous problems than after heterogeneous problems; and this difference was the same under short and long presentation durations. Recall that these strategy sequential difficulty effects have been interpreted as resulting from depletion of memory resources following problems for which the better strategy is more difficult to determine (Schneider & Anderson, 2010; Uittenhove & Lemaire, 2012, 2013a; Uittenhove et al., 2013). This suggests that longer problem presentation did not help participants to recover from this depletion. Note that Uittenhove et al. (2013) found that strategy sequential difficulty effects are influenced by timing parameters, as these effects decreased with increased response–stimulus intervals. Together with the present findings, this suggests that the cognitive system recovers from depletion of processing resources after executing a strategy not during problem encoding but after responding on the previous problem and before encoding the next problem.

In sum, increasing presentation duration helped young and older adults to select the better strategy on each homogeneous and heterogeneous problems more often, when current problems were preceded by either homogeneous or heterogeneous problems. These findings are consistent with the hypothesis that longer duration presentation enhanced encoding processes on current problems for both age groups and for all problems. It is also possible that other strategy selection processes (e.g. inhibition of the just-executed strategy, activation of procedures within each available strategy) may have benefitted from longer duration presentation.

Effects of presentation duration on strategy execution were not as large as on strategy selection. Actually, both young and older adults executed strategy equally efficiently under short and long presentation duration. This suggests that, as assumed by all computational models of strategies, once a problem has been encoded and a strategy selected, encoding processes do not influence execution processes. And this seems to be the case in both young and older adults and for both homogeneous and heterogeneous problems.

One surprising findings in this experiment was that sequence poorer strategy effects (i.e. better performance when problems are estimated by the better strategy as compared to when problems are estimated by the poorer strategy) were found in the short presentation duration condition but not in the long presentation duration condition. That is, young and older adults showed poorer strategy effects after better strategy problems but not after poorer strategy problems only under short presentation duration condition. Under long presentation duration condition, participants showed poorer strategy effects when the preceding problems were estimated both with the poorer and with the better strategy. This was surprising, as we expected that participants would prepare themselves after poorer strategy problems more under the long than under the short presentation duration condition. It is possible that longer problem presentation led participants to focus more on encoding problems, thereby loosing benefits of preparation after poorer strategy problems. It is also possible that, under long presentation duration condition, participants processed poorer strategy problems after poorer strategy problems and after better strategy problems the same way, without benefiting from preparing themselves to process current poorer strategy problems following poorer strategy problems. This led them to be longer to estimate poorer strategy problems than better strategy problems when these problems followed either poorer strategy problems or better strategy problems. In contrast, in the short presentation condition, the benefits of preparation were not absorbed by longer encoding. Like in previous studies (e.g. Hinault et al., 2014; Hinault, Lemaire, & Phillips, 2016; Hinault, Lemaire, & Touron, 2016; Lemaire & Hinault, 2014), participants prepared themselves to process poorer strategy problems after solving poorer strategy problems. This preparation enabled them to be more efficient and to suffer less from having to execute the required poorer strategy when the better strategy is most easily available.

The present results were found in a specific domain (arithmetic problem-solving) and in a specific task (computational estimation). However, strategic variations underlying participants' performance and age-related changes in performance have been found in many other arithmetic tasks (Uittenhove & Lemaire, 2015) and in a wide variety of domains and tasks (see Lemaire, 2015, for an overview). The role of problem presentation duration may generalise to other tasks and domains where strategic variations have been found. To take just one example, both young and older adults have been found to use several encoding strategies in episodic memory tasks (e.g. Bouazzaoui et al., 2010; Hertzog, Price, & Dunlosky, 2012; Kuhlmann & Touron, 2012). Just how long each word of a list is presented may affect how often participants use a mental imagery strategy (i.e. making mental images with each word) and how salient mental images are (hence memory performance, as a consequence).

Whatever the cognitive domain, the present findings of how often the better strategy is selected on each problem as a function of problem presentation durations emphasises the importance of augmenting existing theories of strategy selection. Formal models of strategy choices, like SSL model (Rieskamp & Otto, 2006), SCADS model* (Siegler & Araya, 2005), ACT-R model (Lovett & Anderson, 1996), or RCCL model (Lovett & Schunn, 1999) did not specify how efficient encoding affects strategy selection and strategy execution. They could be revised to include assumptions specifying how quality of encoding the problem contributes to better strategy selection, and how different levels of problem encoding influences strategy selection mechanisms in both young and older adults. This could be done by including assumptions regarding precision, quality, and accuracy of crucial problem representation as a key mechanism by which efficient encoding processes can lead to high levels of better strategy selection and strategy execution in young and older adults.

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